**Machine Learning for Data Science 1, 2025-2026**
**Homework 6 Report**

# Explainable AI

Denis Nedić

*dn98183@student.uni-lj.si*

## 1. Model and LIME Setup

To understand the behavior of the black-box model, a Gradient Boosting Regressor was trained on the provided dataset. The categorical feature $x_3$ was properly handled using one-hot encoding prior to training to avoid introducing false ordinality assumptions. The Local Interpretable Model-Agnostic Explanations (LIME) framework was then implemented from scratch. For the surrogate model, Lasso regression was employed to inherently perform feature selection, yielding sparser and more interpretable local explanations.

## 2. Local Explanations & Feature Importance

The custom LIME algorithm was evaluated on the four specified instances. Across all instances, the explanation assigned a significant positive weight to $x_1$ (ranging from 2.02 to 3.14), identifying it as the overwhelmingly dominant predictive feature in these local neighborhoods.

Thanks to the Lasso regularization, the weight for $x_2$ was reduced to exactly 0.00 in all cases, and $x_3$ yielded either exactly 0.00 or a negligible contribution ($\sim 0.02$). This sparse explanation is faithful to the true behavior of the black-box model. For example, comparing Instance 1 ($x_3 = 1$) and Instance 3 ($x_3 = 2$), the black-box predictions are identical (1.7449), confirming that $x_3$ has essentially no local effect. Similarly, modifying $x_2$ between Instance 3 and Instance 4 changed the black-box prediction by a mere 0.0085, verifying LIME's conclusion that $x_2$ is locally irrelevant.

Lastly, comparing the Black-Box and LIME predictions demonstrates good local fidelity (e.g., 1.74 vs 1.80 for Instance 1). However, the gap in Instance 2 (0.77 vs 1.11) indicates that the true decision surface is highly non-linear in that specific region.

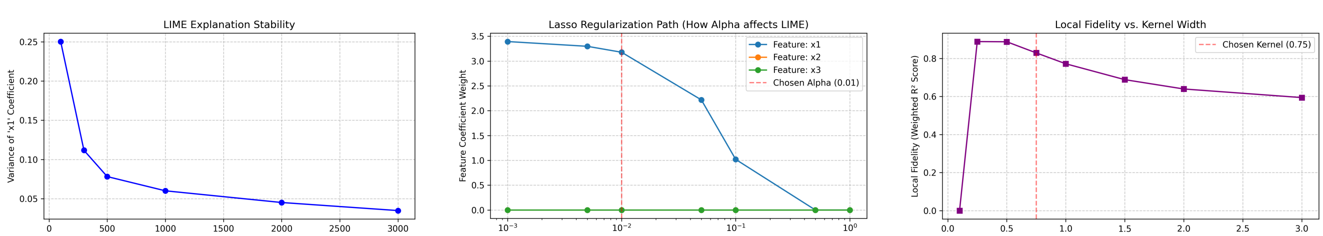
## 3. Implementation Choices & Justification

To ensure the robustness of the LIME explanations, a comprehensive sensitivity analysis was conducted on the core hyperparameters.

**Neighborhood Size ($N$):** The stability of the explanation relies on drawing sufficient samples. Figure 1(a) demonstrates the variance of the $x_1$ coefficient across multiple runs for varying $N$. The variance stabilizes significantly around $N = 1000$. Hence, $N = 1000$ was selected as the optimal trade-off between computational efficiency and explanation reliability.

**Lasso Regularization ($\alpha$):** The regularization parameter dictates the sparsity of the explanation. Figure 1(b) plots the coefficient paths. Increasing $\alpha$ shrinks the weights towards zero. An $\alpha = 0.01$ was chosen because it successfully eliminates noise (forcing irrelevant features to zero) while preserving the valid, dominant contribution of $x_1$.

**Kernel Width:** The exponential kernel width defines the locality of the neighborhood. Figure 1(c) illustrates the local fidelity (measured via weighted $R^2$ score) against different kernel widths. A width of 0.75 yields a peak fidelity score, ensuring the surrogate model accurately mimics the black-box within the relevant local region without overfitting to immediate noise or underfitting by incorporating too much of the global surface.




**(a)** Explanation Stability ($N$) **(b)** Lasso Reg. Path ($\alpha$) **(c)** Local Fidelity (Kernel)

**Figure 1.** Parameter justification analysis for the LIME implementation.